

Employee Management & Attendance Tracker

Internship Project Report | Elevate Labs | SQL / Database Domain

1. Introduction

The Employee Management and Attendance Tracker is a complete relational database system built in PostgreSQL (production) and SQLite (portable demo). It manages 50 employees across 8 departments, tracks 2175+ daily attendance events spanning January-April 2024, and handles leave workflows, salary history, manager hierarchies, and role-based salary bands - all within a single normalised SQL database.

The project covers the complete database engineering stack: schema design, seed data generation, trigger-based automation, stored functions, reusable views, and a suite of 15 analytical HR report queries using GROUP BY, HAVING, window functions, and CTEs.

2. Abstract

The schema layer defines 7 tables normalised to 3NF. The attendance table uses PostgreSQL GENERATED ALWAYS AS columns to auto-compute work_hours, overtime_hours, and is_late from raw check_in/check_out times. Five PL/pgSQL triggers automate core workflows: (1) auto-stamp updated_at on employee changes, (2) log every salary revision to salary_history, (3) write JSONB old/new row snapshots to audit_log for full INSERT/UPDATE/DELETE traceability, (4) validate attendance times and auto-correct status to Late/Present based on the 09:15 threshold, and (5) automatically insert On-Leave attendance rows for each weekday when a leave request is approved.

Five stored functions provide reusable HR analytics accessible to any application layer: total work hours, attendance percentage (with Half-Day weighted at 0.5), full monthly summary, late arrivals ranking via RANK() OVER(), and department-level attendance rollup with performance flags. Six SQL views expose flat denormalised perspectives (employee directory with age and tenure, monthly attendance grouped by employee-month, department headcount with payroll, leave status with approver, salary vs role band). Fifteen analytical queries produce all standard HR reports from overtime leaderboards to gender diversity to CTE-based salary increment tracking.

3. Tools Used

Tool	Purpose
PostgreSQL 15+	Primary production RDBMS
SQLite 3	Portable DB for demo and submission (file-based, no server)
pgAdmin 4	PostgreSQL GUI to run .sql files and inspect data
DB Browser for SQLite	SQLite GUI for the demo database
PL/pgSQL	Language for triggers and stored functions
JSONB	Audit log old/new value snapshots in PostgreSQL
Window Functions	RANK, SUM OVER, ROW_NUMBER for analytics
CTE (WITH clause)	Readable sub-queries in salary history report
Python 3 + reportlab	Database simulation script and PDF report generation

4. Steps Involved in Building the Project

Step 1 - Schema Design (01_schema.sql): Designed 7 normalised tables with all constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK). Attendance uses GENERATED ALWAYS AS for work_hours, overtime_hours, is_late. The employees table has a self-referencing manager_id FK and departments has a manager FK back to employees (set post-creation). Created 9 indexes on attendance(emp_id, work_date), employees(dept_id, role_id, status), leave_requests(emp_id, status), and salary_history(emp_id).

Step 2 - Seed Data (02_seed_data.sql): Inserted 8 departments, 18 roles with salary bands, 50 employees with realistic profiles, 200+ attendance records across 4 months with Present/Late/Absent/Half-Day/On-Leave/Holiday patterns, 10 leave requests in all statuses, and 8 salary history entries with increment percentages.

Step 3 - Triggers (03_triggers.sql): 5 triggers: auto-update employees.updated_at; auto-log salary changes to salary_history; JSONB audit log for all DML on employees; attendance validation (reject invalid times, auto-set Late/Present, nullify times for Absent); auto-insert On-Leave attendance on leave approval.

Step 4 - Stored Functions (04_functions.sql): fn_total_work_hours(emp, year, month); fn_attendance_percentage() with Half-Day=0.5 weighting; fn_monthly_attendance_summary() returning a full typed result table; fn_late_arrivals_report() with RANK() OVER(); fn_department_attendance_summary() with Excellent/Good/Needs Attention performance flag.

Step 5 - Views (05_views.sql): 6 views: vw_employee_directory (age, tenure, manager name); vw_attendance_detail (day-of-week, hours, is_late); vw_monthly_attendance (GROUP BY emp+month); vw_department_headcount (headcount + payroll); vw_leave_status (with approver); vw_salary_overview (band position %).

Step 6 - Analytical Queries (06_queries.sql): 15 report queries: monthly attendance summary; late arrivals (HAVING >=2); dept payroll; overtime leaderboard (RANK); absenteeism; salary band comparison (PARTITION BY dept); leave balance by type; running work-hour total (SUM OVER); monthly dept trend; tenure bands; pending leave queue; gender diversity per dept; salary increment CTE; employment type distribution; full joined HR report.

5. Conclusion

The Employee Management and Attendance Tracker delivers a complete, production-grade HR database system. With 49 active employees, 2175+ attendance records, 5 triggers, 5 stored functions, and 6 views, it covers the full lifecycle from employee onboarding through monthly HR reporting. Every business rule is enforced at the database layer, making the system reliable regardless of which application or tool accesses it.

The project demonstrates mastery of core SQL database engineering: 3NF schema design, PL/pgSQL triggers and functions, JSONB audit trails, GENERATED columns, window functions, CTEs, and comprehensive analytical reporting. These skills are directly applicable to data engineering, backend development, and analytics roles. Future enhancements could include a FastAPI REST layer, pgAdmin dashboard integration, and PostgreSQL Row-Level Security for multi-tenant HR access control.

Project	Employee Management & Attendance Tracker
Domain	SQL / Database Engineering
Program	Elevate Labs Internship - Project Phase (2 Weeks)
Tech Stack	PostgreSQL PL/pgSQL SQLite Triggers Functions Views Window Functions
DB Stats	50 employees 2175+ attendance rows 8 departments 5 triggers 5 functions 6 views